| experiment family | key result (as measured) | sample | test / correction | evidence | verdict |
|---|---|---|---|---|---|
| **Beacon A10: persistent reuse under information asymmetry** | `fas 20/20; direct 0/20; D-noact 0/20; D-flat 20/20 (first action gather\_oak\_log 20/20)` | n=20 x 4 conds | McNemar exact: p=2e-6 (vs direct, noact); p=1.0 (vs D-flat) | L3 behavior | ✓ **relational reuse supported; ablation=full** |
| **Write-back isolation (A1_B1 vs A1_B2)** | `operation:/change: nodes: 0 vs 13 (identical experience, single switch)` | n=1 lineage (seed 1, bitwise) | deterministic isolation (both completions 5/5 @11) | L1 repro. | ✓ **experience -> new structure, not injected** |
| **Cross-context re-activation (G0/G1/G2)** | `marked re-lit 4/12 vs 0/8 vs 0/0 (a>=0.05)` | n=5 seeds | deterministic; vs flat-memory control | L1/L2 | ✓ **episodic structure carries re-activation: access layer** |
| **Functional consumption: ledger + event framework (C, D/C20e/C20g)** | `completion 13 vs 16 ticks; first action flip 5/5; real-failure receipts @15 vs @17` | n=5 determinations | deterministic channel isolation | L3 (assembly) | ✓ **stored experience changes how (not whether) it acts** |
| **JCG: joint convergence gain (core incremental value)** | `JCG>0 in 200/200 paired seeds; >=2-input MRR@1 = 1.0 (prior replication 0.970)` | n=50 x 4 conds | deterministic + prior 0.970 replication | L2 mechanism | ✓ **multi-input convergence; mechanism runs** |
| **dJCG vs flat retrieval (preregistered, C-1..C-4)** | `median dJCG -0.40/-0.40/-0.78/-0.78; reversals FAS>B2: 0/200` | n=50 paired x 4 | reverse-direction 1-sided Wilcoxon + Holm p=1.5e-9; r=0.615 (prereg direction FAS>B2: p~1) | L2 mechanism | ✗ **H3 rejected: no value over flat** |
| **Shared-hub topology + 84-cell parameter sweep** | `FAS 30/30 vs B2 0/30 (w=0.5); sweep: FAS better 11/84 (w<=0.5, d<=1); flat better 9/84 (w>=0.7, d>=1); tie 64` | n=30 seeds x 85 cells | deterministic; design frozen before execution | L2 mechanism | △ **advantage region: weak edges x shallow paths** |
| **Relation directionality (diagnosis, F vs B projection)** | `rho(score,act) 0.325 -> 0.645 (I3); reach2 0 -> 9-12; dilution 0.833->0.475` | n=10 matched x 60 runs | Wilcoxon 21 tests: p=0.001953; r=0.886; 10/10 seeds | L2 mechanism | ✓* **direction semantics=limit; production untouched** |
| **Explicit resource routing (LLM decision head)** | `T4 info-competition 19/20 vs 15/20; raw p=0.046 -> Bonferroni p~1; T3 slower; tokens 2.1-3.7x` | n=20 matched x 520 runs | Wilcoxon; 280 tests Bonferroni: T4 all p~1 | L3 exploratory | △ **directional only; no robust value** |
| **Zero-prior starvation (P0 vs P1/P2)** | `P0 0/25 with +29 nodes; P1=P2 25/25 @median 4 ticks` | n=25 episodes | deterministic; direct replay diagnosis | L3 boundary | △ **scaffolding is precondition; locks** |

Verdicts: ✓ supported as stated; ✗ hypothesis tested and rejected; △ boundary / directional only (✓* = mechanism diagnosis, production graph unmodified).
Effects are per-row incommensurable metrics; only the direction verdict is comparable across rows. Every number is read from the archived run outputs (beacon_v1/analysis_final.json, core_incremental_value/statistics.csv, core_routing_v2 campaign statistics, relation-direction campaign; the prior JCG replication is research_audit/mechanism_falsification).